

6.3.3 Decision Management Process

6.3.3.1 Purpose

The purpose of the Decision Management Process is to select the most beneficial course of action when alternative solutions exist. It provides a structured, transparent approach to decision-making during the system life cycle, regardless of the decision's nature or source, to achieve specified, desirable, or optimized outcomes.

6.3.3.2 Outcomes

As a result of successful implementation of the Decision Management Process:

- a) A decision-making strategy is defined, including decision categories, prioritization, and responsible parties.
 - b) Alternative courses of action are identified and analyzed against established criteria.
 - c) A preferred course of action is selected and approved.
 - d) Decisions, supporting rationale, assumptions, and outcomes are captured, tracked, and reported to relevant stakeholders.
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6.3.3.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.3.3.3.1 Decision Planning

This activity establishes the decision-making strategy, identifies decision needs, and determines responsible parties.

- 6.3.3.3.1.1 Define a decision-making strategy, including:
 - Decision categories (e.g., technical, managerial, contractual)
 - Prioritization schemes
 - Responsible authorities for each category.
- 6.3.3.3.1.2 Identify relevant parties and stakeholders to draw on

appropriate experience and domain knowledge.

- 6.3.3.3.1.3 Identify the circumstances and triggers that require a decision, ensuring that issues, opportunities, or changes are captured in a timely manner.
- 6.3.3.3.1.4 Record, categorize, and promptly report problems or opportunities and possible alternative actions for resolution.

NOTE: Proper categorization and logging of decision requests ensure timely routing to the appropriate decision authorities.

6.3.3.3.2 Decision Analysis

This activity evaluates alternative courses of action against defined criteria to select the most beneficial outcome.

- 6.3.3.3.2.1 Select and declare a decision-making strategy for each decision situation, including analysis methods (e.g., cost-benefit, trade-off, risk assessment).
 - 6.3.3.3.2.2 Identify desired outcomes and define measurable success criteria, including cost, schedule, performance, risk, quality, and compliance factors as appropriate.
 - 6.3.3.3.2.3 Identify and document feasible alternative courses of action.
 - 6.3.3.3.2.4 Evaluate alternatives using the selected strategy, performing trade-off analyses to determine their relative merits.
 - 6.3.3.3.2.5 Select and document the preferred course of action, including rationale, assumptions, and any constraints or dependencies.
 - 6.3.3.3.2.6 Obtain appropriate authorization or approval for the selected decision.
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6.3.3.3.3 Decision Tracking

This activity monitors and reports decision outcomes to ensure effectiveness and support organizational learning.

- 6.3.3.3.3.1 Record and maintain a decision log, including problems, opportunities, alternatives, selected actions, rationale, and disposition, in accordance with organizational procedures or agreements.
- 6.3.3.3.3.2 Track decision implementation to verify that problems are resolved, adverse trends are reversed, and opportunities are realized.
- 6.3.3.3.3.3 Periodically evaluate the impact and effectiveness of decisions against defined success criteria.
- 6.3.3.3.3.4 Report decision outcomes and any follow-up actions to

relevant stakeholders and authorities.

- 6.3.3.3.5 Retain decision records for auditing, traceability, and organizational learning.

Cross-reference:

- Configuration Management Process (6.3.4) for controlling decisions that affect baselined items.
- Measurement Process (6.3.2) for tracking effectiveness metrics.
- Review (7.2.6) and Audit (7.2.7) processes for assessing decision implementation and compliance.